

---

# Sentinel Crowd

AI early warning for crowd crushes using existing CCTV cameras.

Project guide: everything the team needs to understand the project and present it to judges.

|                   |                                                                                                                                              |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Event</b>      | CuriousPARC 2026, Theme 3: Computer Vision, Behaviour Analysis and Scene Understanding                                                       |
| <b>What it is</b> | A web app that watches a camera view, counts people zone by zone, reads how the crowd is moving, and warns staff before it becomes dangerous |
| <b>Code</b>       | <a href="https://github.com/ananyac9820/sentinel-crowd">github.com/ananyac9820/sentinel-crowd</a>                                            |
| <b>Runs on</b>    | Any laptop with Chrome or Edge. No server, no API keys, no special hardware.                                                                 |
| <b>Guide date</b> | 5 October 2026                                                                                                                               |

**How to read this guide.** Sections 1 to 4 explain the idea, what is new about it, and how it works. Sections 5 to 8 walk through every screen, the two modes and every rule. Sections 9 to 13 cover the technology, privacy and limits. Sections 14 and 15 are your presentation script and answers to likely judge questions.

# 1. The problem

Crowd crushes and stampedes happen again and again at Indian railway stations, temples, melas and festivals. They rarely start suddenly. Pressure builds over several minutes, usually at a narrow point such as a staircase, a footbridge or a gate. The people on the ground cannot see it, because each person only sees the few people around them.

Most of these places already have CCTV cameras. But a few people watch many screens, and nobody can judge crowd density and movement on twenty screens at once. By the time someone notices, it is often too late to open a gate or stop people entering.

**Problem in one line: the danger is visible on camera minutes before a crush, but nobody is measuring it.**

# 2. Our solution

Sentinel Crowd turns an ordinary camera view into a control-room screen. It does five things:

1. **Counts people** in each zone of the camera view. Zones are a 4 by 3 grid (A1 to C4) by default, or named areas that staff draw themselves, such as "Staircase".
2. **Reads how the crowd moves:** the direction of flow in each zone, people moving against each other (counter-flow), and chaotic pushing (turbulence).
3. **Predicts** how many seconds remain before a filling zone reaches crush density.
4. **Raises plain-English alerts** with a clear action, such as "Open exit gate 2 and halt platform entry now", and explains which rule fired.
5. **Lets staff try an action** in the simulation and see on a station map where the crowd goes next.

**One-line pitch: Sentinel Crowd reads the crowd, not just the count. It warns before a zone is full, predicts when it will become dangerous, explains why, and tells staff which gate to open, using cameras that are already installed.**

# 3. What makes it new

Detecting and counting people is not new on its own. What is new is what we do with the detections. Use these points when judges ask about novelty.

| Idea                             | What it means                                                                                                                                          | Why it matters                                                                                               |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Crowd turbulence index</b>    | Crowd pressure = density multiplied by how unevenly people move. This idea comes from research on real crowd disasters (Helbing and colleagues, 2007). | Chaotic pushing in a dense crowd is a recognised sign that a crush is starting. Most tools only count heads. |
| <b>Counter-flow detection</b>    | Finds two streams of people moving opposite ways along the same line, for example up and down a staircase.                                             | Opposing flows at stairs and gates are a common trigger of crushes.                                          |
| <b>Time-to-critical forecast</b> | Projects each busy zone's growth to show a countdown: "B3 critical in about 14 seconds".                                                               | Turns detection into prediction. Staff get time to act.                                                      |

| Idea                                         | What it means                                                                                                                           | Why it matters                                                                                     |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| <b>Measured early-warning lead time</b>      | The app records how many seconds before a zone reached CRITICAL the first warning fired. In our scenario: 14 seconds.                   | Proves the early warning actually comes early, with a number.                                      |
| <b>What-if interventions and station map</b> | Operators can open a gate, pause entry or redirect people, and see the crowd respond and move to other areas on a 4-camera station map. | Shows that actions move risk rather than remove it, for example pausing entry fills the concourse. |
| <b>Explainable, actionable alerts</b>        | Every alert says which rule fired with its numbers, and names a specific action and exit gate.                                          | Police and station staff must be able to justify closing a gate.                                   |
| <b>Private and low-cost</b>                  | Runs in a browser on existing cameras. Video never leaves the laptop. No faces recognised.                                              | Realistic for Indian stations, temples and melas.                                                  |

## 4. How it works: the flow

Every half second, the app runs this loop. Simulation mode and Live mode share the same loop; only steps 1 to 3 differ.

| Step                       | What happens                                                                                                                                  | Where in the code                       |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|
| <b>1. Camera</b>           | Video comes from the simulated platform, an uploaded MP4, or a webcam.                                                                        | SimulatedFeed.jsx, useLiveDetection.jsx |
| <b>2. Detect people</b>    | Live: the COCO-SSD model finds every person twice per second. Simulation: the scripted scenario provides counts and movement.                 | detector.js, simulation.js              |
| <b>3. Track movement</b>   | Live: each person is matched to where they were half a second ago, giving a speed and direction. Tracks are anonymous and last a few seconds. | tracker.js                              |
| <b>4. Count per zone</b>   | People and their movement are grouped into zones by position.                                                                                 | zones.js                                |
| <b>5. Smooth</b>           | Counts are averaged (1.5 s in Simulation, 3 s in Live) so flicker does not cause false alarms.                                                | risk.js                                 |
| <b>6. Check rules</b>      | Density, surge, forecast, turbulence and counter-flow for every zone.                                                                         | risk.js                                 |
| <b>7. Overall risk</b>     | Worst zone's level, raised one level if any zone has a surge, turbulence or counter-flow.                                                     | risk.js                                 |
| <b>8. Alert and notify</b> | New alerts appear with time, zone, severity, message and reason. Serious alerts sound an alarm and send a simulated SMS.                      | risk.js, alarm.js, Dashboard.jsx        |

**Everything happens inside the browser tab. Nothing is sent to any server.**

## 5. The screens

### 5.1 Landing page

Product name, one sentence, a **live preview** of the real dashboard running the simulation, an "Open dashboard" button, "How it works" in four steps, and links to the Terms and Privacy pages.

**Sentinel Crowd**

Sentinel Crowd counts people in each part of a CCTV view and warns station or event staff when an area is becoming dangerously crowded.

Built for railway platforms, temples and festivals, where crushes build up over minutes before anyone on the ground can see them.

[Open dashboard](#)

**LIVE PREVIEW** Simulation scenario, Platform 2

**OVERALL RISK**  
**WARNING** Congestion. Divert people away from the flagged zones now.

| CRITICAL IN | PEOPLE | BUSIEST ZONE | RISK SCORE | SURGE | MOTION |
|-------------|--------|--------------|------------|-------|--------|
| B3 ~11s     | 79     | B3 12        | 60 / 100   | B3    | Calm   |

**ALERTS** 4 Newest first. Select an alert to see why.

- 01:12:40 **WATCH** C3 WATCH threshold  
Zone C3 getting busy: 8 people. Keep a stew...
- 01:12:38 **WATCH** B2 WATCH threshold  
Zone B2 getting busy: 8 people. Keep a stew...
- 01:12:35 **WARNING** B3 Forecast  
Zone B3 on course to reach crush density in ...
- 01:12:34 **WARNING** B3 Rapid build-up  
Zone B3 density rising fast: up 85% in 20s. ...

**HOW IT WORKS**

01 **Connect a camera.** Use an existing CCTV feed or a recorded clip. The video is processed in the browser and never leaves the device.

02 **Count people in each zone.** The view is split into a 4x3 grid. A person detection model counts the people in each zone twice a second.

03 **Read the crowd, not just the count.** A zone is flagged when it is too dense, filling too fast, or when people are being pushed in different directions (turbulence) or moving against each other (counter-flow). Its growth is projected forward to

04 **Tell the operator what to do.** Each alert names the zone and a suggested action, such as opening a specific exit gate. Staff can try an action and see on the station map where the crowd goes next.

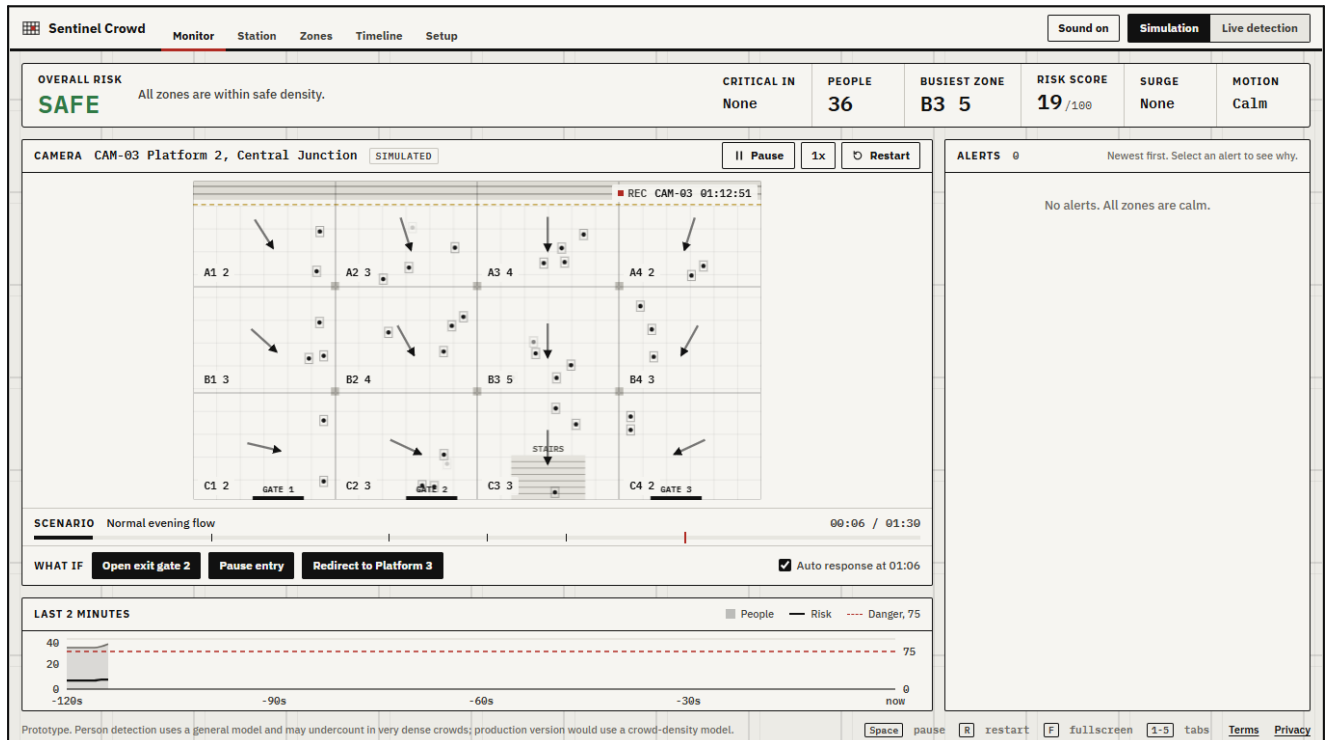
Landing page. The preview is the real dashboard running live, not a picture.

### 5.2 Layout of the dashboard

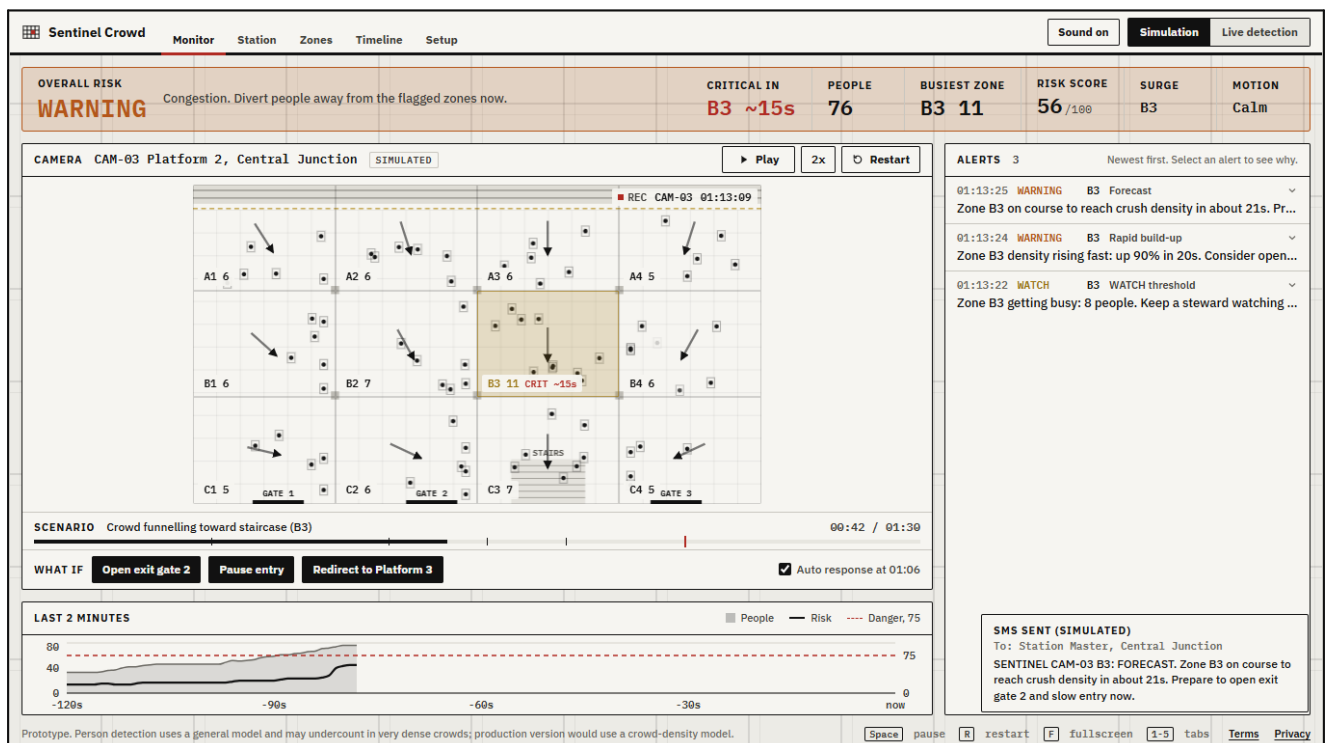
A top bar with five tabs and the Simulation / Live switch, then a status bar that is visible on every tab, so a CRITICAL is never hidden.

| Part                | What it shows                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Status bar</b>   | Overall risk in large letters with a short instruction, then: <b>Critical in</b> (forecast countdown), people, busiest zone, risk score out of 100, zones with a surge, and zones with turbulence or counter-flow (Motion). Tinted at WATCH and WARNING, solid red at CRITICAL, with a red page border. |
| <b>Monitor tab</b>  | Camera view with zone grid and movement arrows, scenario progress, what-if buttons, a small 2-minute chart, and the alerts list.                                                                                                                                                                        |
| <b>Station tab</b>  | Map of four cameras coloured by risk, with arrows for where actions send people.                                                                                                                                                                                                                        |
| <b>Zones tab</b>    | Zone map and a table with count, level, trend, forecast, turbulence, counter-flow, direction and exit gate for every zone.                                                                                                                                                                              |
| <b>Timeline tab</b> | Full chart with turbulence, early-warning lead time and run summary, alert history with filters, incident report, CSV exports, and SMS log.                                                                                                                                                             |
| <b>Setup tab</b>    | Alert thresholds, notification settings (alarm sound, simulated SMS), and the zone editor.                                                                                                                                                                                                              |
| <b>Shortcuts</b>    | Space pause or play, R restart, F fullscreen, 1 to 5 switch tabs. Sound on or off is in the top bar.                                                                                                                                                                                                    |

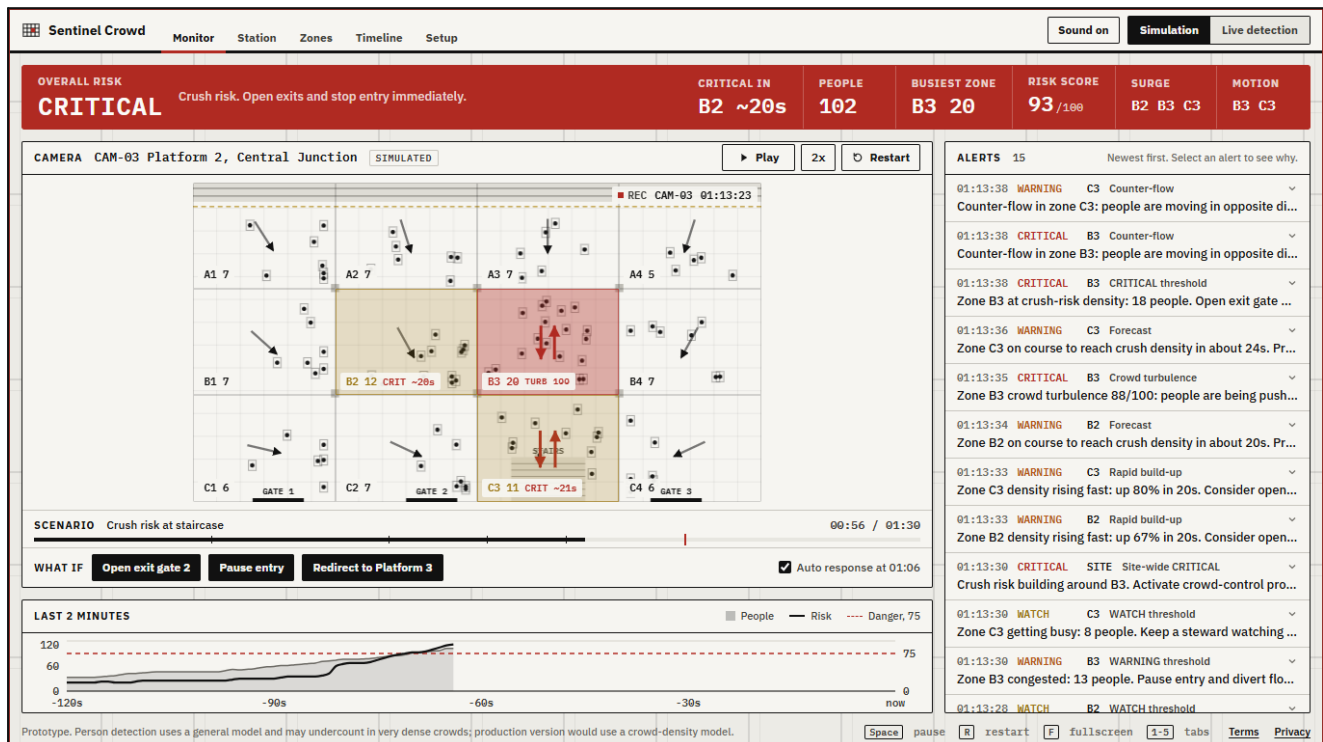
## 5.3 Monitor tab at each risk level



SAFE: normal evening flow. Arrows show everyone drifting toward the staircase.



WARNING at 0:42: B3 next to the staircase is filling fast. The status bar shows "Critical in B3 ~15s" and the forecast alert has fired.



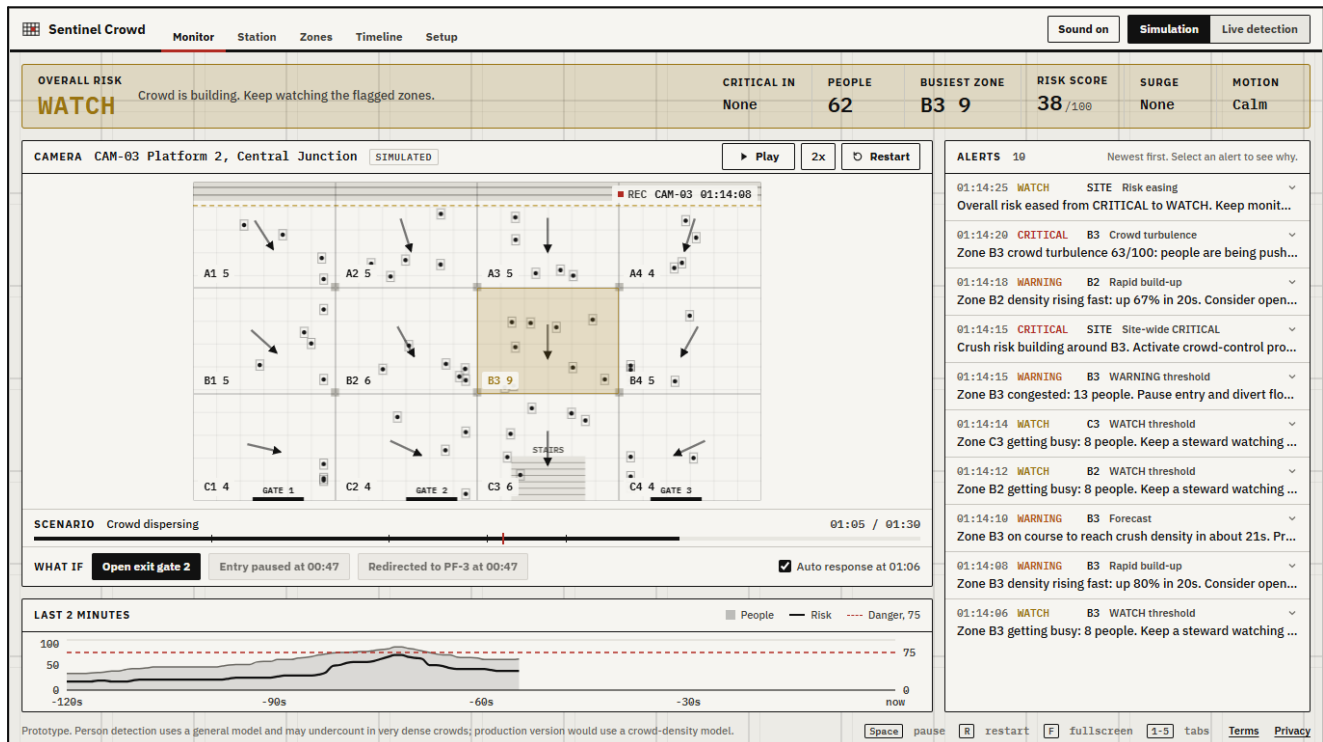
CRITICAL at 0:56: solid red status bar and border. Red double arrows in B3 and C3 show counter-flow; B3 shows turbulence 100. Alerts include crowd turbulence and counter-flow.

## Reading the camera view

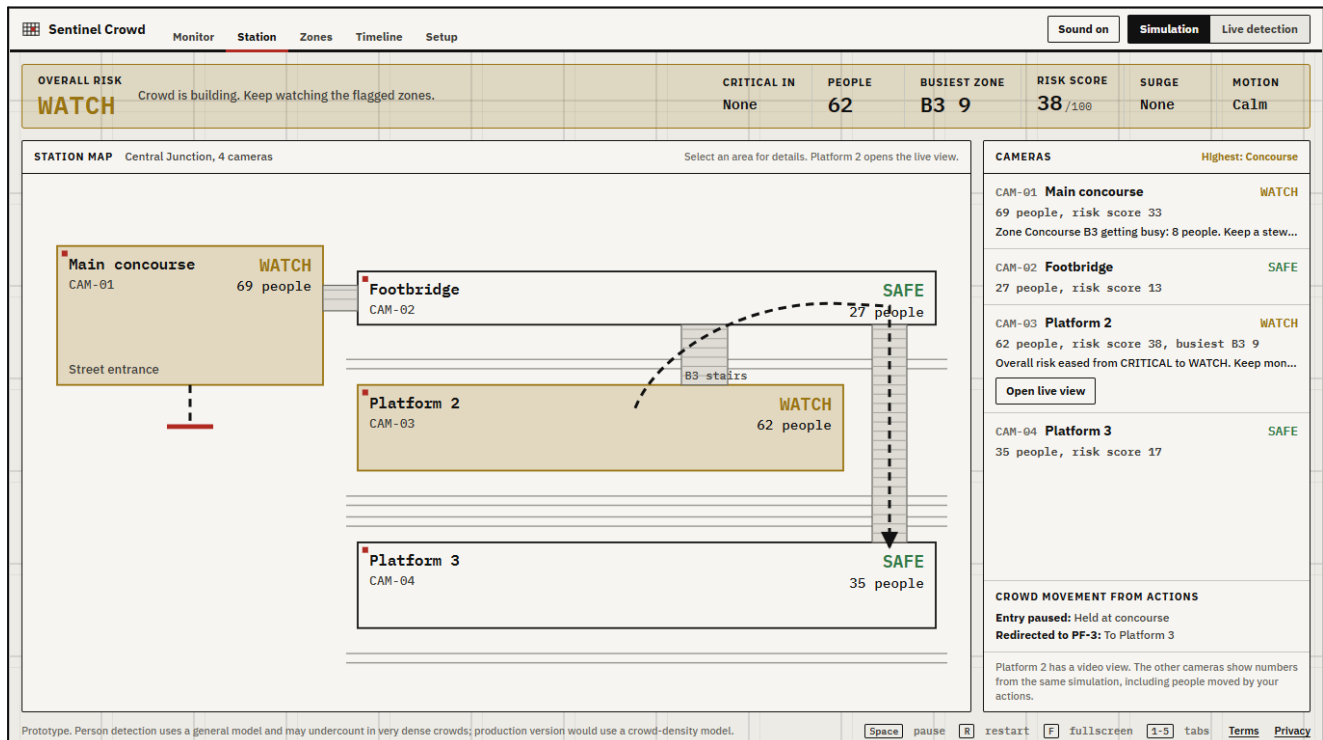
- **Zone colour:** green or none = SAFE, yellow = WATCH, orange = WARNING, red = CRITICAL.
- **Label:** zone name, people count, and a tag: **CRIT ~14s** (forecast), **TURB 94** (turbulence) or **SURGE**.
- **Black arrow:** the direction people in that zone are moving. Longer means faster.
- **Red double arrow:** counter-flow, people moving both ways.

## 5.4 What-if actions and the station map

In Simulation mode, the What if bar has three actions: **Open exit gate 2**, **Pause entry** and **Redirect to Platform 3**. Press one at any time and the crowd responds from that moment. The Station tab shows where the people went.

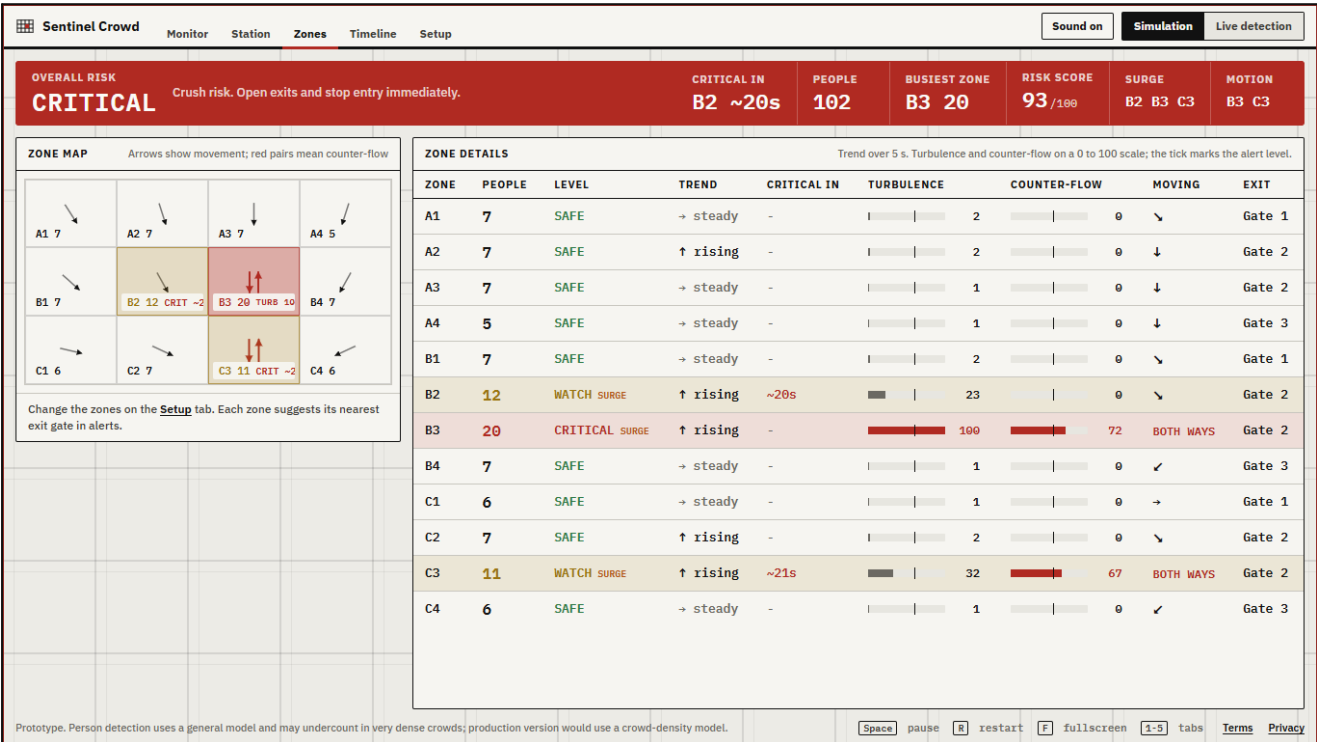


Pause entry and Redirect pressed at 0:47. B3 never reaches crush density; risk falls to WATCH by 1:05.



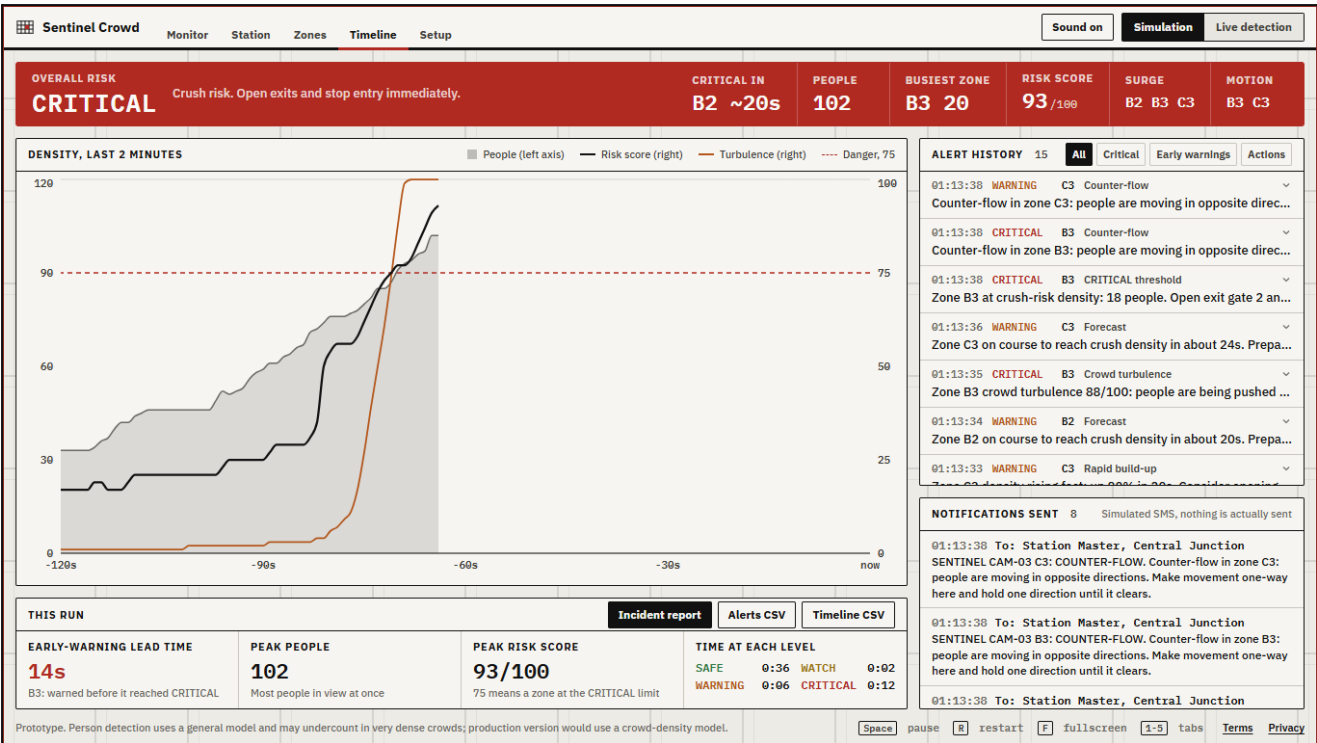
Station map at the same moment: the concourse rises to WATCH because people are held outside, and an arrow shows people moving to Platform 3.

5.5 Zones tab



Zones tab: every zone's numbers in one table. Bars show turbulence and counter-flow on a 0 to 100 scale; the tick marks the alert level.

5.6 Timeline tab and incident report



Timeline tab: full chart with the turbulence line, early-warning lead time (14 s), time at each level, alert history with filters and the notification log.

[Print or save as PDF](#)

## Sentinel Crowd incident report

Central Junction - CAM-03 Platform 2 - Simulation scenario

### SUMMARY

|                         |                                                                         |
|-------------------------|-------------------------------------------------------------------------|
| Period                  | 05/10/2026, 01:11:10 to 05/10/2026, 01:12:42                            |
| Highest overall risk    | <b>CRITICAL</b>                                                         |
| Time at each level      | SAFE 0m 42s   WATCH 0m 14s   WARNING 0m 07s   CRITICAL 0m 29s           |
| Peak people in view     | 112                                                                     |
| Peak risk score         | 100 / 100                                                               |
| Early-warning lead time | B3: first early warning 14s before it reached CRITICAL                  |
| Alerts raised           | level 7, surge 3, forecast 3, site 3, turbulence 2, counter 2, action 1 |
| Actions taken           | 01:12:17 Gate 2 opened                                                  |

### ALERT LOG (OLDEST FIRST)

| Time     | Severity | Zone | Alert                                                                                                                                                                               | Reason                                                                                                                                                                                                                                                |
|----------|----------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01:11:47 | WATCH    | B3   | <b>WATCH threshold</b><br>Zone B3 getting busy: 8 people. Keep a steward watching this area.                                                                                        | Density rule: 8 people in B3 is at or above the WATCH threshold of 8. Counts are averaged over 1.5s to ignore flicker.                                                                                                                                |
| 01:11:49 | WARNING  | B3   | <b>Rapid build-up</b><br>Zone B3 density rising fast: up 80% in 20s. Consider opening exit gate 2.                                                                                  | Surge rule: B3 went from 5 to 9 people (+80%) in 20s, above the trigger of +35% and at least +4 people. A surge raises overall risk by one level.                                                                                                     |
| 01:11:51 | WARNING  | B3   | <b>Forecast</b><br>Zone B3 on course to reach crush density in about 21s. Prepare to open exit gate 2 and slow entry now.                                                           | Forecast rule: B3 grew from 7 to 11 people over the last 10s (about +0.4 people per second). At that rate it crosses the CRITICAL threshold of 18 in about 21s. Forecast alerts fire when the projection is under 25s.                                |
| 01:11:53 | WATCH    | B2   | <b>WATCH threshold</b><br>Zone B2 getting busy: 8 people. Keep a steward watching this area.                                                                                        | Density rule: 8 people in B2 is at or above the WATCH threshold of 8. Counts are averaged over 1.5s to ignore flicker.                                                                                                                                |
| 01:11:55 | WATCH    | C3   | <b>WATCH threshold</b><br>Zone C3 getting busy: 8 people. Keep a steward watching this area.                                                                                        | Density rule: 8 people in C3 is at or above the WATCH threshold of 8. Counts are averaged over 1.5s to ignore flicker.                                                                                                                                |
| 01:11:56 | WARNING  | B3   | <b>WARNING threshold</b><br>Zone B3 congested: 13 people. Pause entry and divert flow towards A3.                                                                                   | Density rule: 13 people in B3 is at or above the WARNING threshold of 13. Counts are averaged over 1.5s to ignore flicker.                                                                                                                            |
| 01:11:56 | CRITICAL | SITE | <b>Site-wide CRITICAL</b><br>Crush risk building around B3. Activate crowd-control protocol: open gate 2, stop announcements drawing people to the platform, dispatch officers.     | Overall rule: worst zone is WARNING (B3), and a surge raises it by one level, giving CRITICAL.                                                                                                                                                        |
| 01:11:59 | WARNING  | B2   | <b>Rapid build-up</b><br>Zone B2 density rising fast: up 75% in 20s. Consider opening exit gate 2.                                                                                  | Surge rule: B2 went from 6 to 11 people (+75%) in 20s, above the trigger of +35% and at least +4 people. A surge raises overall risk by one level.                                                                                                    |
| 01:11:59 | WARNING  | C3   | <b>Rapid build-up</b><br>Zone C3 density rising fast: up 80% in 20s. Consider opening exit gate 2.                                                                                  | Surge rule: C3 went from 5 to 9 people (+80%) in 20s, above the trigger of +35% and at least +4 people. A surge raises overall risk by one level.                                                                                                     |
| 01:11:59 | WARNING  | B2   | <b>Forecast</b><br>Zone B2 on course to reach crush density in about 22s. Prepare to open exit gate 2 and slow entry now.                                                           | Forecast rule: B2 grew from 7 to 11 people over the last 10s (about +0.3 people per second). At that rate it crosses the CRITICAL threshold of 18 in about 22s. Forecast alerts fire when the projection is under 25s.                                |
| 01:12:01 | CRITICAL | B3   | <b>Crowd turbulence</b><br>Zone B3 crowd turbulence 94/100: people are being pushed in different directions. This often comes just before a crush. Stop entry and open exit gate 2. | Turbulence rule: crowd pressure is density multiplied by how unevenly people are moving (Helbing et al., 2007). B3 is at 89% of the CRITICAL count and its movement spread is 0.051 frame heights per second, giving 94/100, above the trigger of 60. |

Incident report: summary (highest risk, time at each level, peak people, lead time, actions) and the full alert log with reasons. Use the browser's Save as PDF to keep it.

**Sentinel Crowd**

Monitor

Station

Zones

Timeline

Setup

Sound on

Simulation

Live detection

OVERALL RISK

CRITICAL

Crush risk. Open exits and stop entry immediately.

CRITICAL IN

B2 ~20s

PEOPLE

104

BUSIEST ZONE

B3 20

RISK SCORE

93 / 100

SURGE

B2 B3 C3

MOTION

B3 C3

ALERT THRESHOLDS

Simulation

Reset

WATCH from

8

people per zone

WARNING from

13

people per zone

CRITICAL from

18

people per zone

Surge: rise of

35

percent

Surge: at least

4

people

Surge: within

20

seconds

Turbulence alert at

60

out of 100

Counter-flow alert at

55

out of 100

How risk is computed

Each zone gets a level from its density. A zone larger or smaller

NOTIFICATIONS

Test alarm

☒ Alarm sound for CRITICAL and early-warning alerts

☒ Simulated SMS to

Station Master, Central Junction

Nothing is actually sent. A real deployment would connect an SMS gateway here.

ZONES

Using the 4x3 grid

Station preset

4x3 grid

Clear

Drag on the view to draw a zone. Click a zone to select it. Avoid overlapping zones.

A1

A2

A3

A4

B1

B2

B3

B4

C1

C2

C3

C4

GATE 1

GATE 2

GATE 3

A1

A1 Gate 1

A2

A2 Gate 2

A3

A3 Gate 2

A4

A4 Gate 3

B1

B1 Gate 1

B2

B2 Gate 2

B3

B3 Gate 2

B4

B4 Gate 3

C1

C1 Gate 1

C2

C2 Gate 2

C3

C3 Gate 2

C4

C4 Gate 3

Apply zones

Discard

Applying restarts monitoring. Limits are scaled to each zone's size, so a big zone needs more people to be flagged.

Prototype. Person detection uses a general alarm and may undercount in very dense crowds; production version would use a crowd-density model.

Space

pause

R restart

F fullscreen

1-5 tabs

Terms

Privacy

Monitor

Station

Zones

Timeline

Setup

Sound on

Simulation

Live detection

OVERALL RISK

WARNING

Congestion. Divert people away from the flagged zones now.

CRITICAL IN

~13s Stair ...

PEOPLE

91

BUSIEST ZONE

25 West pla...

RISK SCORE

62 / 100

SURGE

West platform Sta...

MOTION

Calm

CAMERA

CAM-03 Platform 2, Central Junction

SIMULATED

Play

2x

Restart

SCENARIO

Train arrives: passengers pushing up the stairs against the crowd

00:52 / 01:30

WHAT IF

Open exit gate 2

Pause entry

Redirect to Platform 3

☒ Auto response at 01:06

LAST 2 MINUTES

People

Risk

Danger, 75

ALERTS 7

Newest first. Select an alert to see why.

01:15:07 WATCH West platform Rapid build-up

Zone West platform density rising fast: up 42% in 20s. Co...

01:15:05 WATCH Staircase WATCH threshold

Zone Staircase getting busy: 9 people. Keep a steward wa...

01:15:02 WARNING Stair approach Forecast

Zone Stair approach on course to reach crush density in a...

01:14:56 WATCH Stair approach WATCH threshold

Zone Stair approach getting busy: 16 people. Keep a stew...

01:14:54 WATCH Stair approach Rapid build-up

Zone Stair approach density rising fast: up 46% in 20s. C...

01:14:41 WATCH Platform edge Rapid build-up

Zone Platform edge density rising fast: up 50% in 20s. Co...

01:14:37 WATCH West platform Rapid build-up

Zone West platform density rising fast: up 44% in 20s. Co...

Prototype. Person detection uses a general model and may undercount in very dense crowds; production version would use a crowd-density model.
 

Space

pause

R restart

F fullscreen

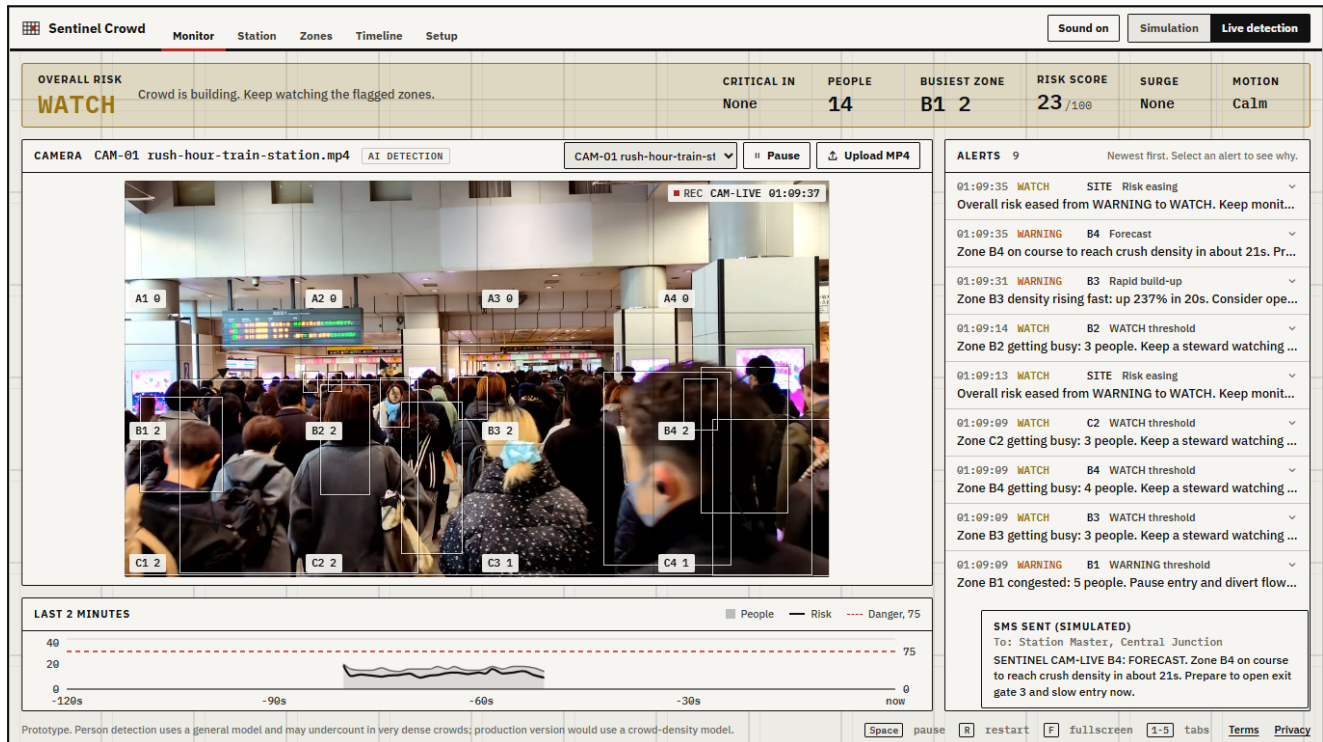
1-5 tabs

Terms

Privacy

Page 10

## 5.8 Live detection on real footage



Live detection on a rush-hour station clip. Each detected person has a box; counts and movement feed the same rules. Very dense crowds are undercounted, which we state openly.

### Other pages

- **Privacy page:** video never leaves the device, no faces are recognised, people are followed only anonymously for a few seconds to measure movement, and nothing is stored.
- **Terms page:** a prototype, not a certified safety system; staff remain responsible for decisions.
- **Error screen:** if anything ever crashes, a "Reload dashboard" message appears instead of a blank page.

## 6. The two modes

### 6.1 Simulation mode (default, use this on stage)

A scripted 90-second scenario on Platform 2 of a busy junction. It needs no camera and no internet, and runs exactly the same way every time. Press R to restart from SAFE. With "Auto response" ticked, Gate 2 opens by itself at 1:06.

| Time         | What happens on screen                                                                           |
|--------------|--------------------------------------------------------------------------------------------------|
| 0:00         | Normal evening flow. Everything <b>SAFE</b> .                                                    |
| 0:18         | Train delay announced. The platform slowly fills.                                                |
| 0:37         | B3, next to the staircase, passes 8 people. Level <b>WATCH</b> .                                 |
| 0:39         | Surge alert: "Zone B3 density rising fast". Overall risk jumps to <b>WARNING</b> .               |
| 0:40         | Forecast alert: B3 on course to reach crush density in about 20 to 25 seconds. Countdown starts. |
| 0:45         | Overall <b>CRITICAL</b> . Status bar turns red.                                                  |
| 0:46         | A train arrives and passengers push up the stairs against the crowd.                             |
| 0:49 to 0:51 | Crowd turbulence alert in B3, then counter-flow alerts in B3 and C3 (red double arrows).         |
| 0:53         | B3 actually reaches the critical count of 18, 14 seconds after the first early warning.          |
| 1:06         | Auto response: Gate 2 opens. (Or press a what-if button earlier and see a different outcome.)    |
| 1:14 to 1:26 | Crowd disperses. Risk eases to <b>WATCH</b> , then <b>SAFE</b> .                                 |

### What-if results you can show

| Action at about 0:47                  | What happens                                                                                  |
|---------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Nothing (untick Auto response)</b> | B3 keeps rising past 25 people and stays <b>CRITICAL</b> .                                    |
| <b>Open exit gate 2</b>               | Platform empties quickly through gate 2; the footbridge gets busier for a while.              |
| <b>Pause entry</b>                    | B3 never reaches crush density, but the concourse fills to <b>WATCH</b> : risk moves outside. |
| <b>Redirect to Platform 3</b>         | B3 peaks lower; Platform 3 gets busier.                                                       |

### 6.2 Live detection mode

1. Click **Live detection** in the top bar.
2. The first time, the browser downloads the AI model (about 67 MB, 15 to 45 seconds). It is then saved on the laptop, so later it loads in seconds even without internet.
3. When it says "AI model ready", click **Upload MP4** or **Use webcam**. Choosing a new camera starts a fresh analysis.
4. Twice per second, every person is boxed, tracked for movement, counted into a zone, and the same rules run on the real data.

Tips: high-angle CCTV-style clips work best. Sample clips are in the demo-videos folder on the laptop (the Indian platform and rush-hour clips work well; the stairs clip is a close-up and is not useful). The

webcam view is mirrored like a video call. Webcams only work on https or localhost. To connect a real CCTV camera today, use OBS Studio's Virtual Camera or a phone webcam app such as DroidCam.

## 7. The risk rules in detail

All rules are deliberately simple so they can be explained to anyone in one sentence. Defaults are shown for Simulation / Live.

| Rule                  | What it checks                                                                                                                                                                          | Default (Simulation / Live)                                 |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Density levels</b> | People in a zone compared with three limits. Limits are for a standard grid cell and scale with zone size, so this is really density.                                                   | WATCH 8 / 3, WARNING 13 / 5, CRITICAL 18 / 8 people         |
| <b>Surge</b>          | Count rose by at least a percentage AND at least a number of people within a time window, and is still rising.                                                                          | +35% and +4 people in 20 s / +50% and +2 people in 20 s     |
| <b>Forecast</b>       | Straight line fitted to the zone's count over the last 10 s, projected to the CRITICAL limit. Only for zones at WATCH or above that are still rising.                                   | Shown if under 60 s; alert if under 25 s                    |
| <b>Turbulence</b>     | Crowd pressure = (people / CRITICAL limit) x (movement spread / reference spread) squared, shown on a 0 to 100 scale. Movement spread is how differently people in the zone are moving. | Alert at 60 or more, held for 1.5 s, zone at WATCH or above |
| <b>Counter-flow</b>   | Find the main line of movement; if people move both ways along it, the index rises toward 100 (equal streams). Random jostling scores 0.                                                | Alert at 55 or more, held for 1.5 s, zone at WATCH or above |
| <b>Overall risk</b>   | Worst zone's level, raised one level while any zone has a surge, turbulence or counter-flow.                                                                                            | Same in both modes                                          |
| <b>Risk score</b>     | 0 to 100. 75 means a zone is at its CRITICAL limit. +10 while a surge, turbulence or counter-flow is active.                                                                            | Same in both modes                                          |
| <b>Lead time</b>      | Seconds between a zone's first early warning (surge, forecast, turbulence, counter-flow) and the moment it reached CRITICAL.                                                            | Shown in the Timeline tab and report                        |

### Protections against false alarms

- Counts are averaged over 1.5 s (Simulation) or 3 s (Live).
- A zone must drop at least one whole person below a limit before the same alert can fire again, and the same level never repeats within 30 s.
- Surge, forecast, turbulence and counter-flow alerts have a 30 s cooldown per zone. A zone that is emptying out is never called a surge.
- In Live mode, people only report movement after being tracked in three consecutive frames, and small movement jitter (0.02 frame heights per second) is ignored.
- The overall level must stay lower for 3 s before it is lowered, so it does not flicker.

## 8. Technology used

| Tool                               | What we used it for                                                       | Why we chose it                             |
|------------------------------------|---------------------------------------------------------------------------|---------------------------------------------|
| <b>React 19</b>                    | The user interface, built from components                                 | Fast to build, widely known                 |
| <b>Vite 8</b>                      | Development server and production build                                   | Very fast; outputs a simple static website  |
| <b>Tailwind CSS 4</b>              | Styling and the paper control-room theme                                  | Quick, consistent design                    |
| <b>TensorFlow.js 4</b>             | Runs the AI model in the browser using the graphics card (WebGL)          | No server needed; video stays on the device |
| <b>COCO-SSD (MobileNet v2)</b>     | Pre-trained model that finds people in each frame                         | Free, well-tested, real time on a laptop    |
| <b>Our own tracker</b>             | Matches people between frames to measure speed and direction (tracker.js) | Small, anonymous, no extra model            |
| <b>Recharts 3</b>                  | Density and risk charts                                                   | Simple React charts                         |
| <b>HTML Canvas and SVG</b>         | Animated simulated platform, flow arrows, station map                     | Smooth graphics with no video files         |
| <b>Web Audio API</b>               | Alarm tones                                                               | No sound files needed                       |
| <b>IndexedDB (browser storage)</b> | Saves the AI model after first download                                   | Works offline on demo day                   |
| <b>GitHub, Vercel or Netlify</b>   | Code hosting and free website hosting                                     | One-click deploy                            |

Language: JavaScript (React JSX). There is no backend, no database and no API key.

## 9. How the code is organised

| File                                  | What it does                                                                                                                          |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>src/lib/risk.js</b>                | The brain: smoothing, density levels, surge, forecast, turbulence, counter-flow, overall risk, lead time, alert messages and reasons. |
| <b>src/lib/zones.js</b>               | Zone layouts (grid and custom), exit gates, neighbours, grouping people into zones, the counter-flow measure.                         |
| <b>src/lib/tracker.js</b>             | Live mode person tracker for speed and direction.                                                                                     |
| <b>src/lib/simulation.js</b>          | The scripted scenario, motion script, what-if actions, and the other three station cameras.                                           |
| <b>src/lib/detector.js</b>            | Loads COCO-SSD and saves it in the browser for offline use.                                                                           |
| <b>src/lib/report.js, alarm.js</b>    | Incident report, CSV exports; alarm tones.                                                                                            |
| <b>src/hooks/useLiveDetection.jsx</b> | Upload or webcam, the twice-per-second detection loop, loading and error screens.                                                     |
| <b>src/components/Dashboard.jsx</b>   | Tabs, simulation clock, interventions, notifications and keyboard shortcuts.                                                          |

| File             | What it does                                                                                                                                                    |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other components | StatusCard, CameraPanel, ZoneGrid, SimulatedFeed, AlertsFeed, DensityChart, ZoneTable, StationMap, TimelineView, SettingsPanel, ZoneEditor, Landing, LegalPage. |

## 10. Privacy and safety

- All processing happens on the laptop inside the browser. Video files and camera streams are never uploaded.
- The model only draws boxes around people. It does not recognise faces or estimate age or gender.
- To measure movement, each box is followed for a few seconds using only its position on screen. These tracks have no identity and are discarded straight away.
- Nothing is stored except the public AI model files in the browser cache. Alerts disappear when the page is reloaded. SMS messages are simulated and never sent.
- The app supports decisions. Staff and authorities remain responsible for every action.

## 11. Honest limitations

- The general-purpose detector undercounts very dense crowds, where people hide behind each other. A production version would use a crowd-density model trained on Indian footage.
- Movement is estimated from detections twice per second, so speeds and directions are approximate, and tracking gets harder in very packed scenes.
- Camera angle matters. Overhead or high-angle views work best.
- Real CCTV cameras usually stream in a format browsers cannot open (RTSP). Today we connect them through OBS Virtual Camera; a production version would add a small relay server.
- The forecast is a straight-line projection. When a crowd speeds up, the real time can be shorter than predicted.
- The Simulation scenario, other station cameras and SMS messages are scripted or simulated.

## 12. How to run and deploy

```
npm install
npm run dev
```

Then open <http://localhost:5173> in Chrome or Edge. Stop it with Ctrl+C. To deploy on Vercel: sign in with GitHub, Add New, Project, import sentinel-crowd, keep the Vite settings, Deploy. Open Live detection once on the deployed link before demo day so the model is saved there.

## 13. What we would build next

- A crowd-density model trained on Indian stations and festivals, for accurate counts in packed crowds.
- A small relay server to connect many real CCTV cameras directly, with one dashboard for a whole station.
- Real SMS and wireless alerts to staff, and integration with station announcement systems.
- Learning normal patterns for each station and time of day, to flag unusual build-ups earlier.

## 14. Demo script (about 90 seconds)

Press R first so the scenario starts from SAFE. Use 1x speed. Keep Sound on so judges hear the alarm.

| Time              | Say this                                                                                                                                                                                      | Do this                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>0 to 10 s</b>  | "Stampedes at Indian stations and temples happen because nobody sees the danger building. Sentinel Crowd watches existing CCTV and warns before the crush."                                   | Landing page, then Open dashboard, press F.                                |
| <b>10 to 25 s</b> | "This is Platform 2 at rush hour. Each zone is counted, and the arrows show which way people are moving."                                                                                     | Point at the zone grid and arrows.                                         |
| <b>25 to 40 s</b> | "B3 by the staircase is filling fast. We warn early and forecast it: crush density in about 15 seconds."                                                                                      | Point at the surge alert and Critical in countdown.                        |
| <b>40 to 55 s</b> | "A train arrives and people push up the stairs against the crowd. We detect that counter-flow, and the turbulence: chaotic pushing that research links to crushes. Every alert explains why." | Point at the red double arrows and TURB tag; click one alert.              |
| <b>55 to 70 s</b> | "The operator can act. If we pause entry, the platform recovers, but the station map shows the concourse filling. Risk moves; it doesn't disappear."                                          | Press Pause entry; press 2 for the Station tab.                            |
| <b>70 to 90 s</b> | "Afterwards, the incident report shows we warned 14 seconds before the zone became critical. And in Live mode the same logic runs on real video, in the browser, with no faces recognised."   | Press 4 for Timeline, open Incident report; then switch to Live detection. |

## 15. Questions judges may ask

| Question                                                       | Short answer                                                                                                                                                                                                                                           |
|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Is person detection your innovation?</b>                    | No. Detection is a standard building block. Our contribution is reading crowd behaviour: turbulence, counter-flow, a time-to-critical forecast, measured lead time, and what-if actions with a station view.                                           |
| <b>What is crowd turbulence?</b>                               | When a crowd is dense and people start moving in different directions, pressure waves form. Research on the 2006 Hajj disaster (Helbing and colleagues) measured this as density times the variance of movement. We compute a simple version per zone. |
| <b>How do you measure movement without identifying people?</b> | Each box is matched to the nearest box half a second earlier. That gives a speed and direction. There is no face or identity, and tracks are discarded after a few seconds.                                                                            |
| <b>How accurate is it?</b>                                     | The general model undercounts very dense crowds, which we state openly. The rules use trends and relative changes, which still work when counts are low. A crowd-density model can replace the detector without changing the rest.                     |
| <b>Why rules instead of a black-box risk model?</b>            | Operators must trust and justify actions. Simple rules with written reasons are auditable, and limits can be tuned per site.                                                                                                                           |
| <b>How does the forecast work?</b>                             | A straight line fitted to the zone's count over the last 10 seconds, extended to the critical limit. It only runs for busy zones that are still rising.                                                                                                |
| <b>Can it work with real CCTV?</b>                             | Yes, through OBS Virtual Camera today. A full deployment adds a small relay that converts camera streams for the browser.                                                                                                                              |
| <b>What does it cost?</b>                                      | An ordinary laptop, a browser, free open-source software and cameras that are already installed.                                                                                                                                                       |
| <b>What if the internet fails?</b>                             | Simulation needs no internet. Live mode needs it once to download the model; after that it is cached.                                                                                                                                                  |